

Suggested Teaching Guidelines for
Mathematics and Statistics for AI
PGCP-AI August 2026

Duration: 60 Hours: 60 Theory (T)

Objective: To introduce the student to fundamentals of Artificial Intelligence.

Prerequisites: Knowledge of programming language, Computing Fundamentals, and some basic statistical knowledge.

Evaluation method: Theory exam – 80%, Internal Assessment– 20%

List of Course material

Text book:

Artificial Intelligence: A Modern Approach, 4e, Peter Norvig, Stuart J. Russell, Pearson Education, 4th Edition.

Reference Book:

1. Artificial Intelligence by Example, Denis Rothman
2. Artificial Intelligence by Saroj Kaushik

Note:

- ***Each session mentioned is of 2 hours' theory. Faculty can give case studies to students***
- ***Faculties are advised to relate the topics with real world applications.***

I. Fundamentals of AI

Session 1:

Theory:

- Definition and scope of AI
- AI evolution and historical milestones
- Alan Turing's work
- Turing Machine
- Turing Test

Session 2:

Theory:

- Ethics of AI
- Being human in the age of AI
- Responsible AI
- Social, legal, and ethical implications

Sessions 3 & 4:

Theory:

- Structure of AI systems
- Real-world implications of AI
- AI revolution and Current trends in Artificial Intelligence

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Sessions 5 & 6:**Theory:**

- Artificial life
- Learning through interaction
- Emergent behavior
- Rules-based systems
- Expert systems

Sessions 7 & 8:**Theory:**

- Supervised learning
- Unsupervised learning
- Learning models and examples

Session 9:**Theory:**

- Knowledge representation techniques
- Problem-solving approaches
- State-space representation

Sessions 10 & 11:**Theory:**

- Types of search problems
- Classical search methodologies
- Beyond classical search
- Parallel search
- Search engines
- Adversarial search

Sessions 12 & 13:**Theory:**

- Intelligent agents
- Uninformed search
- Constraint satisfaction search
- Combinatorial optimization problems

Session 14:**Theory:**

- Knowledge representation and automation
- Propositional logic
- Predicate logic
- Logic concepts
- Logic programming

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Session 15:**Theory:**

- Inference and resolution for problem solving
- Advanced problem-solving paradigm: Planning
- Introduction to Big Data
- Structured and unstructured data
- Relevance of Big Data in AI
- Data analysis and data analytics
- Applications of Big Data

II. Mathematics for AI**a. Linear Algebra****Session 1**

- Vectors, Definition,
- Scalars, Addition, Scalar Multiplication
- Inner Product (Dot Product), Vector Projection
- Cosine Similarity, Orthogonal Vectors

Sessions 2 & 3

- Normal And Orthonormal Vectors
- Vector Norm, Vector Space
- Linear Combination, Linear Span
- Linear Independence, Basis Vectors

Assignments: Consider the vectors $\{[3, 0, 4], [-1, 0, 7], [2, 9, 11]\}$. Check that the vectors are linearly independent or not?

Session 4

- Linear Independence
- Basis and Rank
- Linear Mappings
- Affine Spaces

Session 5

- Matrices Definition, Addition, Transpose
- Scalar Multiplication, Matrix Multiplication, Matrix Multiplication Properties
- Hadamard Product, Functions, Linear Transformation, Determinant, Identity Matrix,
- Invertible

Assignments:

- $X = [0 \ 1 \ 3]^T$ and $Y = [2 \ 4 \ 0]^T$
- Find $V = \text{Subspace of } X$
 - Find $W = \text{Subspace of } Y$
 - Describe $V \cap W$?

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Session 6

- Matrix and Inverse, Rank, Trace
- Popular Type of Matrices- Symmetric, Diagonal, Orthogonal, Orthonormal
- Positive Definite Matrix
- Matrix Phylogeny
- Matrix Approximation

Sessions 7 & 8

- Eigen Values & Eigenvectors, Concept, Intuition, Significance
- How To Find Principle Component Analysis
- Concept, Properties, Applications
- Singular Value Decomposition

Assignments:

Find the Eigen values and eigenvectors for the matrix

$$\begin{bmatrix} 5 & -10 & -5 \\ 2 & 14 & 2 \\ -4 & -8 & 6 \end{bmatrix}$$

b. Calculus**Session 9 & 10**

- Functions, Scalar Derivative, Definition, Intuition
- Common Rules Of Differentiation, Chain Rule
- Partial Derivatives, Gradient
- Concept, Intuition, Properties
- Directional Derivative

Assignments: Let $f(x,y,z)=xyex^2+z^2-5$ $f(x,y,z)=xyex^2+z^2-5$. Calculate the gradient of f at the point $(1,3,-2)$ and calculate the directional derivative $D_u f$ at the point $(1,3,-2)$ in the direction of the vector $v=(3,-1,4)$.

Session 11

- Gradients of Vector Valued Functions
- Gradient of Matrices
- Useful Identities for Computing Gradients
- Back propagation and Automatic Differentiation
- Linearization and Multivariate Taylor Series

Sessions 12 & 13

- Optimization Using Gradient Descent
- Constrained Optimization and Lagrange Multipliers
- Convex Optimization

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Sessions 14 & 15

- Vector And Matrix Calculus
- How To Find Derivative Of Scalar-Valued function
- Vector-Valued Function With Respect To Scalar, Vector , Four Combinations- Jacobian
- Gradient Algorithms, Local/Global Maxima and Minima
- Saddle Point, Convex Functions
- Gradient Descent Algorithms- Batch, Mini-Batch, Stochastic
- Performance Comparison